SUPERMARKET BILLING SYSTEM

A Python-Based Mini Project

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PYTHON • AUTOMATED BILLING • ITEMIZED RECEIPT



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Real-World Billing Scenario

From Shopping Cart to Final Bill



Scenario

A customer buys multiple items at a supermarket. The cashier needs to enter each item's name, quantity and price, calculate the total amount, apply any applicable discount and generate an itemized bill (receipt).

Example Purchase

Item Name	Quantity	Price (₹)
Apple	2	20.00
Banana	10	5.00
Carrot	6	3.00
Radio	1	500.00
Speaker	1	2500.00

Key Requirements

- Process multiple items in a single transaction
- Apply discounts based on certain conditions (e.g., total amount, item type or promotions)
- Calculate and display subtotal, discount and final total
- Generate an itemized receipt with all details
- Use loops and conditional statements to handle continuous input until billing is complete

Smart Billing for a Better Shopping Experience



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Billing Logic: From Items to Final Amount

How Python Processes Your Shopping Cart

Simple Steps, Smart Billing!

1 Item Input

Get item name, quantity and price from user.

2 Calculate Amount

Multiply quantity × price for each item.

3 Check Discount

If subtotal ≥ ₹1,000 apply 10% discount.

4 Final Amount

Subtract discount from subtotal.

5 Generate Receipt

Show itemized bill, subtotal, discount and total.

Example Calculation

Item	Quantity	Price (₹)	Amount (₹)
Apple	2	20.00	40.00
Banana	10	5.00	50.00
Carrot	6	3.00	18.00
Radio	1	500.00	500.00
Speaker	1	2500.00	2500.00

Bill Summary

Subtotal	**₹3,108.00**
Discount (10%)	**₹310.80**
Final Total	**₹2,797.20**

Accurate Calculation, Happy Customers!



04 Code Walkthrough & Actual Output

From Input to Final Bill – See How It Works

Python + Jupyter Notebook

Simple Code Powerful Results!



Key Code Logic

1 Item Input
Get item name, quantity and price from user (using loop).

2 Calculate Amount
Multiply quantity × price for each item.

3 Discount Condition
If subtotal ≥ ₹1,000 → apply 10% discount (using if-else).

4 Final Calculation
Subtract discount from subtotal to get final total.

5 Print Receipt
Display itemized bill, subtotal, discount and final total.

Python Code (Key Parts)

```
items = []
subtotal = 0

while True:
    name = input("Enter item name (or 'done' to finish): ")
    if name.lower() == "done":
        break

    quantity = int(input("Enter quantity: "))
    price = float(input("Enter price per item: "))
    amount = quantity * price
    items.append([name, quantity, price, amount])
    subtotal += amount

if subtotal >= 1000:
    discount = subtotal * 0.10
else:
    discount = 0

total = subtotal - discount

# Print receipt
print("\n------- SUPERMARKET RECEIPT ------")
print(f"{"Item":15}{"Qty":>5}{"Price":>10}{"Amount":>12}")

for item in items:
    print(f"{item[0]:15}{item[1]:>5}{item[2]:>10.2f}{item[3]:>12.2f}")

print("-"*40)
print(f"Subtotal:        ₹{subtotal:.2f}")
print(f"Discount:        ₹{discount:.2f}")
print(f"Final Total:     ₹{total:.2f}")
print("Thank you for shopping!")
```

Sample Output

```
Enter item name (or 'done' to finish): Apple
Enter quantity:  2
Enter price per item: 20
Enter item name (or 'done' to finish): Banana
Enter quantity: 10
Enter price per item: 5
Enter item name (or 'done' to finish): Carrot
Enter quantity:  6
Enter price per item: 3
Enter item name (or 'done' to finish): Radio
Enter quantity: 1
Enter price per item: 500
Enter item name (or 'done' to finish): Speaker
Enter quantity:  1
Enter price per item: 2500
Enter item name (or 'done' to finish): done

-------- SUPERMARKET RECEIPT -------
Item             Qty      Price        Amount
Apple              2      20.00         40.00
Banana            10       5.00         50.00
Carrot             6       3.00         18.00
Radio              1     500.00        500.00
Speaker            1    2500.00       2500.00
-----------------------------------------------
Subtotal:          ₹3108.00
Discount:          ₹310.80
Final Total:       ₹2797.20

Thank you for shopping!
```

What You See Here

- **User input** for multiple items
- **Itemized receipt** (with quantity, price, amount)
- **Subtotal, discount** and final total
- Clean and readable formatting

Pro Tip

The same logic can be extended to include GST, product categories, barcode scanning and file saving.

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Conclusion & Key Learnings

What We Learned & What's Next

Small Code Big Impact!

Python Supermarket Billing System

Key Learnings

01 Loops & Conditions
Used while, for loops and if-else statements for logical flow and calculations.

02 Lists & Data Handling
Managed multiple items using lists and indexing.

03 User Input & Output
Collected data from user and displayed formatted results (itemized receipt).

04 Real-World Application
Automated billing process with discount logic and receipt generation.

Future Scope

GST Calculation
Add GST based on item category and applicable rate.

Product ID & Barcode Support
Scan barcodes for faster and error-free billing.

Inventory Management
Track stock levels and alert when items are low.

Save Bill to File
Store billing history in a file for future reference.



Thank You!

For Your Time & Attention

Automating Today for a Smarter Tomorrow